

Changyu (Thomason) Zhao

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EDUCATION

University of Wisconsin - Madison

Bachelor of Science in Computer Science (Honors) and Math

Sep. 2021 – May 2025

Madison, Wisconsin

EXPERIENCE

UW - Madison Cybersecurity Operation Center

Cybersecurity Student Intern Analyst

Feb. 2023 – Present

Madison, Wisconsin

- Analyzed daily outbound malicious traffic reports from Palo Alto Firewall and Cisco Secure Endpoint (AMP).
- Traced malicious traffic in Elastic and Palo Alto Panorama to notify security alert to related user or department.
- Conducted security authorization and Identity Access Management (IAM) for access restrictive accounts.

Tencent Spark Early Career Program

Cybersecurity Researcher Intern

Jun. 2021 – Present

Shenzhen, China

- Conducted source code review of the modified Vijos Online Judge (OJ) system, identifying deserialization and sandbox escape vulnerabilities.
- Successfully exploited the identified vulnerabilities, gaining root privileged reverse shell within Docker container.
- Demonstrated privilege escalation to root level on host machine by leveraging CVE-2019-5736 to escape from privileged Docker container.
- Utilized IDA for reverse engineering router firmware, located buffer overflow vulnerability mentioned in CVE-2018-8879.
- Applied Return Oriented Programming (ROP) techniques to establish reverse shell and achieve full control over the router.

PROJECTS

Tenda AC18 AC1900 Router Exploitation | IDA, GDB, QEMU, Python

May 2022 – Aug. 2022

- Exploited unfixed Common Vulnerabilities and Exposures (CVE) in Tenda AC18 router firmware.
- Employed qemu-system to emulate arm32 router firmware on an x86-64 machine, enabling systematic debugging.
- Developed automated stack overflow and command injection detection plug-in for IDA at decompiled code level.
- Successfully achieved Remote Code Execution (RCE) by exploiting vulnerabilities identified by the IDA plug-in.
- Discovered multiple unfixed CVEs: CVE-2018-16334, CVE-2018-5767, CVE-2018-5768, CVE-2018-5770, CVE-2020-15916, CVE-2022-28556, CVE-2022-28557.

Windows Calculator Reverse Engineering | IDA, Windbg, Frida

Jan. 2022 – Apr. 2022

- Exploited the Win10 and Win32 Calculator to manipulate arithmetic operations, resulting in intentionally incorrect results (e.g., $1 + 1 = 3$).
- Conducted both static and dynamic analysis on the calculator binary using IDA and Windbg, unraveling its inner workings.
- Utilized Frida to apply hooks on the calculator's library functions, enabling the hijacking of arbitrary arithmetic operations.

EXTRACURRICULAR ACTIVITIES

r3kapiG CTF Team

Team member

Sep. 2021 – Present

- Achieved a notable accomplishment by securing a Rank 12/1282 worldwide in the DEF CON CTF Finals.
- Actively participated in over 20 48-hour Capture the Flag (CTF) competitions since 2021, with a focus on Binary Exploitation (PWN) and Reverse Engineering challenges.

pwn.college

Independent Learning

Dec. 2021 – Jul. 2022

Tempe, Arizona

- Attained the Yellow Belt and Blue Belt levels by completing all 472 challenges in the Dojo instance, currently ranked 43 out of 2500+ participants.
- Acquired comprehensive skills in various areas of binary exploitation, including program interaction, shellcoding, privilege escalation, reverse engineering, stack corruption, heap corruption, race conditions, and kernel security.

TECHNICAL SKILLS

Languages: C/C++, Java, Python, x86-Assembly JavaScript, HTML/CSS, L^AT_EX

Binary Security Tools/Framework: IDA/Ghidra, GDB/Windbg, Angr, Frida, Elastic Search, Joern

Web Security Tools: Burpsuit, Wireshark, Metasploit Framework, Kali Linux